WORKED SOLUTION

STEP II 1991, Question 7

Pure • Functional Equations

STEP II 1991, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q7

The key idea

A functional equation holds for *all* x and y, so we may differentiate it with respect to one variable while holding the other fixed. That turns it into a differential equation linking g' at two different points. Choosing one of the variables cleverly then pins g' down everywhere, and integrating finishes the job. Throughout we assume, as the question does, that g and f are differentiable.

Solution

The derivative relation

Hold $y > 0$ fixed and differentiate $\mathrm{g}(x) + \mathrm{g}(y) = \mathrm{g}(z)$ with respect to x. Since $\mathrm{g}(y)$ is now a constant, the chain rule gives

$$\mathrm{g}'(x) = \mathrm{g}'(z)\,\frac{\partial z}{\partial x}.$$

By the quotient rule,

$$\frac{\partial z}{\partial x} = \frac{y(x+y+1) - xy}{(x+y+1)^2} = \frac{y^2+y}{(x+y+1)^2},$$

which is the first form. For the second, notice that $z+1$ factorises neatly:

$$z+1 = \frac{xy+x+y+1}{x+y+1} = \frac{(x+1)(y+1)}{x+y+1}, \qquad \text{so} \qquad z(z+1) = \frac{xy(x+1)(y+1)}{(x+y+1)^2}.$$

Dividing by $x(x+1)$ gives exactly $\dfrac{y(y+1)}{(x+y+1)^2}$, so

$$\mathrm{g}'(x) = \frac{y^2+y}{(x+y+1)^2}\,\mathrm{g}'(z) = \frac{z(z+1)}{x(x+1)}\,\mathrm{g}'(z),$$

as required. It is worth writing this in the symmetric form

$$x(x+1)\,\mathrm{g}'(x) = z(z+1)\,\mathrm{g}'(z). \tag{†}$$

Deducing $2g'(1) = (u^2+u)g'(u)$

Ask yourself We want a statement about every u in $(0,1)$. Which variable should we fix, and at what value?

Put $x = 1$ in $(\dagger)$; the left side becomes $2g'(1)$ and $z = \dfrac{y}{y+2}$. As y runs through $(0,\infty)$, $\dfrac{y}{y+2} = 1 - \dfrac{2}{y+2}$ increases continuously from 0 towards 1, so it takes every value u with $0 < u < 1$ (explicitly, $y = \frac{2u}{1-u}$). Hence

$$2g'(1) = (u^2+u)\,g'(u) \qquad \text{for all } 0 < u < 1.$$

Integrating

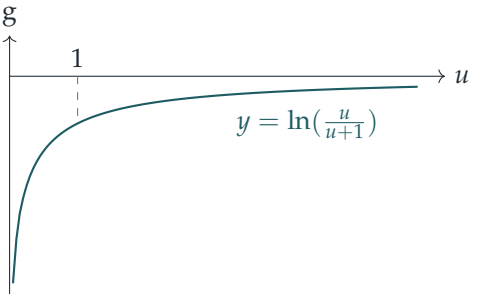
Write $A = 2g'(1)$, a constant. Then for $0 < u < 1$, using partial fractions,

$$g'(u) = \frac{A}{u(u+1)} = A\left(\frac{1}{u} - \frac{1}{u+1}\right),$$

and integrating (everything is positive, so no modulus signs are needed),

$$g(u) = A(\ln u - \ln(u+1)) + B = A\ln\left(\frac{u}{u+1}\right) + B.$$

In fact the formula holds for *every* $u > 0$, not just $u < 1$. For fixed x, $z = \dfrac{xy}{x+y+1}$ increases from 0 towards x as y goes from 0 to ∞, so we can choose y to make $z < 1$. Then $(\dagger)$ and the result above give $x(x+1)g'(x) = z(z+1)g'(z) = A$. So the same integration works on all of $(0,\infty)$, with one constant B, since $(0,\infty)$ is an interval.



(The curve shown is the case $A = 1$, $B = 0$: it is negative, increasing, and tends to 0 as $u \to \infty$.)

Checking $(*)$ and finding B

Substitute $g(u) = A\ln\frac{u}{u+1} + B$. The left-hand side of $(*)$ is

$$g(x) + g(y) = A\ln\frac{xy}{(x+1)(y+1)} + 2B.$$

For the right-hand side, the factorisation of $z+1$ found earlier gives

$$\frac{z}{z+1} = \frac{xy}{x+y+1} \cdot \frac{x+y+1}{(x+1)(y+1)} = \frac{xy}{(x+1)(y+1)},$$

so

$$\mathrm{g}(z) = A \ln \frac{xy}{(x+1)(y+1)} + B.$$

The two sides agree for all $x, y > 0$ if and only if $2B = B$, that is, $B = 0$. So $\mathrm{g}(u) = A \ln \dfrac{u}{u+1}$ satisfies $(*)$.

Can A be determined?

No. The verification above works for *every* value of A, including $A = 0$ (the zero function). Nor does the definition $A = 2\mathrm{g}'(1)$ help: for our solution, $\mathrm{g}'(1) = A(1 - \frac{1}{2}) = \frac{A}{2}$, so $A = 2\mathrm{g}'(1)$ is automatically true and says nothing new. The equation $(*)$ is linear in g, so any multiple of a solution is again a solution, and A is just that scale factor.

The function f

The same method works. Fix $y > 0$ and differentiate $\mathrm{f}(x) + \mathrm{f}(y) = \mathrm{f}(xy)$ with respect to x:

$$\mathrm{f}'(x) = y\,\mathrm{f}'(xy).$$

Ask yourself What is the analogue here of putting $x = 1$?

Putting $x = 1$ gives $\mathrm{f}'(1) = y\,\mathrm{f}'(y)$ for every $y > 0$, so with $C = \mathrm{f}'(1)$,

$$\mathrm{f}'(y) = \frac{C}{y} \qquad \Longrightarrow \qquad \mathrm{f}(y) = C \ln y + D \quad (y > 0).$$

Substituting back, $\mathrm{f}(x) + \mathrm{f}(y) = C\ln(xy) + 2D$ while $\mathrm{f}(xy) = C\ln(xy) + D$, so $D = 0$. (Equivalently, $x = y = 1$ in the equation gives $2\mathrm{f}(1) = \mathrm{f}(1)$, so $\mathrm{f}(1) = 0$.) Hence $\mathrm{f}(x) = C \ln x$, as required, and once again C is arbitrary.

Checks

- Take $x = 1$, $y = 2$: then $z = \frac{1}{2}$ and the two derivative factors are $\frac{6}{16}$ and $\frac{\frac{1}{2}\cdot\frac{3}{2}}{2} = \frac{3}{8}$, which agree.
- With $A = 1$: $\mathrm{g}(1) + \mathrm{g}(2) = \ln\frac{1}{2} + \ln\frac{2}{3} = \ln\frac{1}{3}$, and $\mathrm{g}(\frac{1}{2}) = \ln\frac{1/2}{3/2} = \ln\frac{1}{3}$. ✓
- A computer confirmed the derivative identity by finite differences at 1000 random points, and that $\mathrm{g}(x) + \mathrm{g}(y) - \mathrm{g}(z)$ equals B exactly (so vanishes only for $B = 0$) for several values of A.

Common slips

- Forgetting that $g(y)$ is a constant when differentiating with respect to x, or forgetting the chain-rule factor $\partial z/\partial x$.
- Putting $x = 1$ but not checking that z then really covers all of $(0, 1)$; this is what turns a single equation into a statement for every u.
- Assuming B must be 0 "because constants of integration don't matter": it is the equation $(*)$ that forces $B = 0$, and that needs to be shown.

Beyond STEP

The second equation is Cauchy's logarithmic functional equation: putting $x = e^s$ turns it into $h(s + t) = h(s) + h(t)$. Here we assumed differentiability; in a first analysis course you see that continuity at a single point is already enough to force $h(s) = cs$, whereas without any regularity there are wild non-linear solutions (built using a Hamel basis). The first equation is the same thing in disguise: $w = \ln \frac{u}{u+1}$ transforms $(*)$ into the additive equation. Abbott's *Understanding Analysis* is a good place to read about continuity arguments of this kind.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.